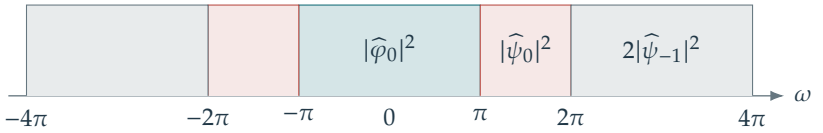
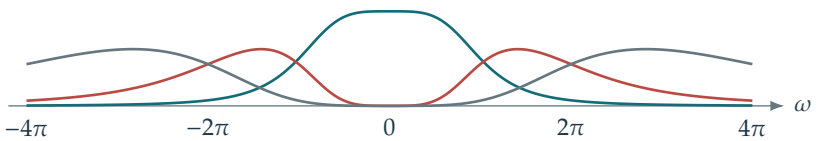
Shannon: sharp bands



Splines: smooth overlap (schematic)



$$2^{-j}|\widehat{\psi}_{j,0}(\omega)|^2 = |\widehat{\psi}(2^j\omega)|^2$$

$$\sum_{j\in\mathbb{Z}} |\widehat{\psi}(2^j\omega)|^2 = 1 \qquad \text{(a.e. } \omega \neq 0)$$